```
#define VRy A1
#define threshold 200

// Motor driver pins
#define IN1 10
#define IN2 9

// LEDs
#define LED_FORWARD 3
#define LED_LEFT 4
#define LED_RIGHT 5

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(LED_FORWARD, OUTPUT);
  pinMode(LED_LEFT, OUTPUT);
  pinMode(LED_RIGHT, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  int y = analogRead(VRy);

  // ------------------------
  // Forward
  // ------------------------
  if (y < 512 - threshold) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    digitalWrite(LED_FORWARD, HIGH);
    digitalWrite(LED_LEFT, LOW);
    digitalWrite(LED_RIGHT, LOW);

    Serial.println("Forward");
  }

  // ------------------------
  // Backward
```

```
  // -------------------------
  else if (y > 512 + threshold) {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);

    digitalWrite(LED_FORWARD, LOW);
    digitalWrite(LED_LEFT, HIGH);
    digitalWrite(LED_RIGHT, HIGH);

    Serial.println("Backward");
  }

  // -------------------------
  // Center (Stop)
  // -------------------------
  else {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);

    digitalWrite(LED_FORWARD, LOW);
    digitalWrite(LED_LEFT, LOW);
    digitalWrite(LED_RIGHT, LOW);
  }

  delay(300);
}
```